Package ‘bulknull’

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Title Assessing Whether a Bulk-Null Result Can Distinguish Cell-State Dilution from Genuine Absence

Version 0.1.0

Description Diagnostics for bulk transcriptomic null results in studies that also have single-cell data. Computes a Dilution Discordance Score, the posterior probability that a non-significant bulk effect reflects dilution of a small subcluster rather than genuine absence, together with a Diagnosability Index quantifying whether the study design has power to make that distinction at all. Includes an applicability gate that refuses bulk-significant inputs.

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Contents

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bulknull *One-Call Wrapper for Bulk-Null Dilution Diagnostics*

Description

Unified interface that runs the applicability gate, computes the Dilution Discordance Score (DDS), and the Diagnosability Index (DI) in a single call. Returns a formatted S3 object summarizing the complete analysis.

Usage

```
bulknull(
  bulk_beta,
  bulk_se,
  bulk_fdr,
  sc_zscore,
  cluster_fraction,
  condition_fraction,
  n_cluster,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass"),
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

`bulk_beta`	numeric; effect size estimate from bulk analysis
`bulk_se`	numeric; standard error of bulk_beta
`bulk_fdr`	numeric; FDR (or p-value) from bulk test
`sc_zscore`	numeric; z-score of signal in target single-cell cluster
`cluster_fraction`	numeric; fraction of total cells in target cluster
`condition_fraction`	numeric; fraction of cells in case condition
`n_cluster`	numeric; number of cells in target cluster
`null_fdr_threshold`	numeric; FDR threshold for applicability gate (default 0.05)
`on_violation`	character; "error", "warn", or "pass" (default "error")
`alpha`	numeric; significance level for DI (default 0.05)
`sided`	character; "one" or "two" for DI computation (default "one")

Details

The bulknull workflow is:

1. Check applicability: bulk FDR > threshold (default 0.05)
2. If applicable, compute DDS (posterior probability of dilution)
3. Compute DI (power to detect dilution)
4. Return formatted S3 object with interpretation bands

The function preserves the full output of dilution_score() and diagnosability_index() internally for inspection if needed.

Value

An S3 object of class "bulknull" containing:

`inputs`	List of input parameters
`applicability`	List with $applicable (logical) and $note (character)
`dds`	DDS (Dilution Discordance Score) value
`mu_dilution`	Expected bulk z-score under dilution hypothesis
`z_bulk`	Observed bulk z-score
`di`	Diagnosability Index value
`verdict`	Interpretive string (e.g., "Ambiguous")
`gate_status`	Character indicating whether input passed the gate

Examples

```
# Example: DPP9 in M8 myeloid cells from IPF cohort
result <- bulknull(
  bulk_beta = -0.084210,
  bulk_se = 0.062774,
  bulk_fdr = 0.449745,
  sc_zscore = 2.077409,
  cluster_fraction = 1332 / 19175,
  condition_fraction = 0.631,
  n_cluster = 1332,
  alpha = 0.05,
  sided = "one"
)

print(result)
```

check_dds_applicability

Check DDS Applicability: Bulk Must Be Null

Description

Validates whether a case is applicable for DDS analysis.

DDS answers the question: "Given a bulk-NULL result and strong single-cell signal, is the null due to dilution or true absence of effect?"

This function enforces the applicability boundary:

- **Applicable:** bulk FDR > null_fdr_threshold (non-significant bulk result)
- **Not Applicable:** bulk FDR <= null_fdr_threshold (significant bulk result)

When not applicable, DDS is mathematically valid but answers the wrong question.

Usage

```
check_dds_applicability(
  bulk_fdr,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass")
)
```

Arguments

`bulk_fdr` Numeric. FDR-adjusted p-value from bulk analysis.

`null_fdr_threshold` Numeric. FDR threshold for bulk to be considered "null". Default: 0.05 (standard significance threshold).

`on_violation` Character. One of:

- "error": Stop with informative error message (strict mode)
- "warn": Issue warning but return results (permissive mode)
- "pass": Return results silently (developer/testing mode)

Value

A list with elements:

- applicable: Logical. TRUE if bulk_fdr > null_fdr_threshold (NULL), FALSE otherwise.
- applicability_note: Character string explaining the status.
- violation_severity: Character. "none", "warning", or "error" depending on on_violation setting.
- bulk_fdr: The input FDR value.
- null_fdr_threshold: The threshold used.
- call: The matched function call.

Examples

```
# Case 1: Bulk is non-significant (FDR = 0.15) - APPLICABLE
check_dds_applicability(bulk_fdr = 0.15, null_fdr_threshold = 0.05,
                        on_violation = "warn")

# Case 2: Bulk is significant (FDR = 0.001) - NOT APPLICABLE
check_dds_applicability(bulk_fdr = 0.001, null_fdr_threshold = 0.05,
                        on_violation = "warn")
```

`diagnosability_index` *Diagnosability Index (DI)*

Description

Compute statistical power to detect dilution under a one-sided or two-sided test.

Usage

```
diagnosability_index(mu_dilution, alpha = 0.05, sided = c("one", "two"))
```

Arguments

`mu_dilution`	Numeric. Expected signal under dilution: mu = (cluster_frac * sc_zscore) / (bulk_se * sqrt(n_eff)).
`alpha`	Numeric. Significance level (default 0.05).
`sided`	Character. "one" for one-sided test (default, recommended for directional alternatives) or "two" for two-sided.

Details

DI estimates the probability of correctly detecting dilution if it truly occurred at the specified effect size (mu_dilution).

One-sided (default, recommended):

- Critical value: z = qnorm(1 - alpha) = 1.644854 for alpha=0.05
- Rationale: Dilution alternative is directional by construction (sign(mu_dilution) = sign(sc_zscore) since cluster_frac > 0)
- Formula: DI = pnorm(mu - 1.644854)

Two-sided (for reference):

- Critical value: z = qnorm(1 - alpha/2) = 1.959964 for alpha=0.05
- Formula: DI = pnorm(mu - 1.959964)

Power categories:

- DI < 0.3: Low power (undetectable at this effect size)
- DI 0.3-0.8: Moderate power (detection possible but unreliable)
- DI > 0.8: High power (reliable detection)

Value

List containing:

- di: Diagnosability Index (power), in range 0 to 1
- critical_value: Critical z-value used
- alpha: Significance level
- sided: Test type ("one" or "two")
- formula: Human-readable formula string

Examples

```
# DPP9/M8 k=5 tissue-only
diagnosability_index(mu_dilution = 0.130536, alpha = 0.05, sided = "one")
# Returns: di = 0.0650 (very low power; dilution undetectable)
```

`dilution_scan` *Vectorized Dilution Scan Across Multiple Genes*

Description

Applies the bulknull framework to multiple genes in a single call, returning a summary table with DDS, DI, and verdict for each gene. Skips bulk-significant genes (FDR <= threshold) and genes with missing single-cell entries.

Usage

```
dilution_scan(
  bulk_deg,
  sc_zscores,
  cluster_fraction,
  condition_fraction,
  n_cluster,
  null_fdr_threshold = 0.05,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

`bulk_deg` data frame; differential expression results with columns `gene` (character), `beta` or `log2fc` (numeric effect size), `se` or `se_log2fc` (numeric standard error), and `fdr` or `padj` (numeric FDR). Additional columns are ignored.

`sc_zscores` named numeric vector; z-scores keyed by gene name. Genes not in this vector are skipped with a note.

`cluster_fraction` numeric; fraction of total cells in target cluster (constant across all genes).

`condition_fraction` numeric; fraction of cells in case condition (constant across all genes).

`n_cluster` numeric; number of cells in target cluster (constant across all genes).

`null_fdr_threshold` numeric; FDR threshold for applicability gate (default 0.05). Genes with FDR <= threshold are marked "FAILED".

`alpha` numeric; significance level for DI (default 0.05).

`sided` character; "one" or "two" for one-sided or two-sided test (default "one").

Details

The function iterates over rows of bulk_deg, skipping:

- Genes with bulk FDR <= null_fdr_threshold (gate fails)
- Genes without an entry in sc_zscores
- Genes with zero variance in the single-cell data (sc_zscore = 0)

For applicable genes, it calls bulknull() internally and extracts key results. Returns all processed genes in output, with NA or "SKIPPED" for missing data.

Value

A data frame with columns:

`gene` Gene name from bulk_deg

`bulk_beta` Effect size from bulk_deg

`bulk_se` Standard error from bulk_deg

`bulk_fdr` FDR from bulk_deg

`sc_zscore` Single-cell z-score (NA if not found)

`mu_dilution` Expected bulk z-score under dilution hypothesis

`dds` Dilution Discordance Score

`di` Diagnosability Index

`dds_interpretation` Categorical interpretation of DDS

`di_interpretation` Categorical interpretation of DI

`verdict` Final interpretive verdict

`gate_status` PASSED or FAILED (applicability gate status)

Examples

```
# Example: scan a small DEG table with mock z-scores
bulk_deg <- data.frame(
  gene = c("GENE1", "GENE2", "GENE3"),
  beta = c(-0.084, 0.020, 0.150),
  se = c(0.063, 0.050, 0.100),
  fdr = c(0.450, 0.05, 0.001)
)
sc_zscores <- c(GENE1 = 2.077, GENE2 = 1.5, GENE3 = 0.5)

result <- dilution_scan(
  bulk_deg = bulk_deg,
  sc_zscores = sc_zscores,
```

```
  cluster_fraction = 0.0695,
  condition_fraction = 0.631,
  n_cluster = 1332
)
print(result)
```

dilution_score *Dilution Discordance Score (DDS)*

Description

Compute the Bayesian posterior probability that an observed bulk-level effect reflects true dilution of a single-cell signal.

Usage

```
dilution_score(
  bulk_beta,
  bulk_se,
  bulk_fdr,
  sc_zscore,
  cluster_fraction,
  condition_fraction,
  n_cluster,
  null_fdr_threshold = 0.05,
  on_violation = c("error", "warn", "pass")
)
```

Arguments

bulk_beta	Numeric. Observed bulk effect size (Hedges' g or log2-fold-change).
bulk_se	Numeric. Standard error of bulk effect size.
bulk_fdr	Numeric. FDR-adjusted p-value from bulk analysis (or raw p-value). Used for validation only; not directly in likelihood.
sc_zscore	Numeric. Single-cell module z-score or effect z-score (e.g., z-score of module enrichment in the subcluster).
cluster_fraction	Numeric. Fraction of total tissue represented by the responsible subcluster (0 < f <= 1).
condition_fraction	Numeric. Fraction of subcluster cells from the condition of interest (e.g., IPF, 0 <= p <= 1).
n_cluster	Integer. Number of cells in the subcluster.
null_fdr_threshold	Numeric. FDR threshold for bulk to be considered "null". Default: 0.05. Only used if bulk_fdr is provided.
on_violation	Character. One of "error", "warn", "pass". Controls behavior when bulk FDR is significant. Default: "warn".

Value

A list with elements:

- dds_score: Posterior probability of dilution (0 to 1).
- z_bulk: Bulk z-score (beta / SE).
- mu_dilution: Expected z under dilution model.
- log_lik_h1: Log-likelihood under dilution model.
- log_lik_h0: Log-likelihood under null model.
- summary: Plain-text interpretation.
- applicable: Logical. TRUE if case is applicable (bulk is null), FALSE if not.
- applicability_note: Character string describing applicability status.
- call: The matched function call.

Examples

```
# DPP9 in M8 (IPF inflammasome macrophages)
dilution_score(
  bulk_beta = -0.0297,
  bulk_se = 0.0595,
  bulk_fdr = 0.821,
  sc_zscore = 2.077,
  cluster_fraction = 1332 / 19175,
  condition_fraction = 0.631,
  n_cluster = 1332
)
```

---

dpp9_m8 *DPP9/M8 Example Dataset*

Description

A list containing measurements for DPP9 (diphosphonucleotidase 9) expression in the M8 (inflammatory macrophages) subcluster from IPF lung tissue.

Usage

```
data(dpp9_m8)
```

Format

A list with 12 elements:

bulk_beta Numeric. Bulk effect size (-0.0297)

bulk_se Numeric. Standard error of bulk effect (0.0595)

bulk_fdr Numeric. FDR-adjusted p-value from meta-analysis (0.821)

sc_zscore Numeric. Single-cell module z-score (2.077)

cluster_fraction Numeric. Fraction of tissue in M8 (0.0695)

condition_fraction Numeric. Fraction of M8 from IPF (0.631)

n_cluster Integer. Number of cells in M8 (1332)

gene Character. Gene name ("DPP9")

cluster Character. Cluster identifier ("M8")

cluster_name Character. Cluster description ("Inflammatory Macrophages")

cohort Character. Disease cohort ("IPF")

description Character. Dataset description

Details

This is a real data example from the bulknull application: a bulk-level null result (non-significant effect) paired with strong single-cell signal in a small subcluster. Used throughout vignettes and examples.

The DPP9/M8 case exemplifies the framework's utility: bulk RNA-seq from IPF lung tissue shows no significant DPP9 effect (FDR = 0.821), but single-cell analysis reveals strong inflammasome-module enrichment in M8. The DDS computes whether the bulk null reflects true absence or dilution of this M8 signal by the dominant other cell types.

Result: DDS ~ 0.48 (ambiguous), suggesting both null and dilution remain plausible. The Diagnosability Index (DI ~ 0.065) reveals that the study design is fundamentally underpowered for dilution detection due to the small M8 fraction (6.95% of lung), independent of effect size.

Examples

```
data(dpp9_m8)
str(dpp9_m8)

# Compute Dilution Discordance Score
result <- dilution_score(
  bulk_beta = dpp9_m8$bulk_beta,
  bulk_se = dpp9_m8$bulk_se,
  bulk_fdr = dpp9_m8$bulk_fdr,
  sc_zscore = dpp9_m8$sc_zscore,
  cluster_fraction = dpp9_m8$cluster_fraction,
  condition_fraction = dpp9_m8$condition_fraction,
  n_cluster = dpp9_m8$n_cluster
)
print(result$dds_score)
```

`plot.bulknull` *Plot DI Landscape for Study Design*

Description

Visualizes the Diagnosability Index as a function of cluster fraction and bulk measurement precision (SE), marking the current study's position. Base graphics only.

Usage

```
## S3 method for class 'bulknull'
plot(
  x,
  sc_zscore = NULL,
  condition_fraction = NULL,
  n_cluster = NULL,
  alpha = 0.05,
  sided = c("one", "two"),
  cluster_fraction_range = c(0.001, 1),
  se_range = c(0.001, 0.5),
  grid_points = 50,
  ...
)
```

Arguments

`x`	A bulknull object
`sc_zscore`	numeric; single-cell z-score (same as input to bulknull)
`condition_fraction`	numeric; case fraction in cohort
`n_cluster`	numeric; cells in target cluster
`alpha`	numeric; significance level (default 0.05)
`sided`	character; "one" or "two" (default "one")
`cluster_fraction_range`	numeric vector of length 2; range for y-axis
`se_range`	numeric vector of length 2; range for x-axis (log scale)
`grid_points`	numeric; number of grid points per dimension (default 50)
`...`	Additional arguments (ignored)

Details

The landscape shows DI as a 2D function of cluster_fraction and bulk_se. Higher DI (>0.8) indicates good power; lower DI (<0.3) indicates underpowered. Contours mark DI = 0.3 and DI = 0.8 boundaries. The study is marked with a red dot at (bulk_se, cluster_fraction) from the input bulknull object.

`print.bulknull` *Print method for bulknull objects*

Description

Displays a formatted verdict block showing inputs, computed statistics, interpretations, and the final verdict.

Usage

```
## S3 method for class 'bulknull'
print(x, ...)
```

Arguments

`x`	A bulknull object
`...`	Additional arguments (ignored)

`print.required_design` *Print method for required_design objects*

Description

Print method for required_design objects

Usage

```
## S3 method for class 'required_design'
print(x, ...)
```

Arguments

`x`	A required_design object
`...`	Additional arguments (ignored)

`required_design` *Required Study Design to Achieve Target Diagnosability Index*

Description

Inverts the diagnosability index formula to find the study parameters needed to achieve a target power (DI). Can solve for either cluster_fraction (at fixed bulk_se) or bulk_se (at fixed cluster_fraction).

Usage

```
required_design(
  target_di,
  sc_zscore,
  condition_fraction,
  n_cluster,
  cluster_fraction,
  bulk_se,
  alpha = 0.05,
  sided = c("one", "two")
)
```

Arguments

`target_di`	numeric; target diagnosability index (0 to 1)
`sc_zscore`	numeric; single-cell z-score for the signal
`condition_fraction`	numeric; fraction of cells in case condition (0 to 1)
`n_cluster`	numeric; number of cells in target cluster
`cluster_fraction`	numeric; fraction of total cells in target cluster
`bulk_se`	numeric; standard error of bulk effect estimate
`alpha`	numeric; significance level (default 0.05)
`sided`	character; "one" or "two" for one-sided or two-sided test (default "one")

Details

The diagnosability index is defined as: DI = Normal_CDF(mu - z_critical) where Normal_CDF is the standard normal CDF and z_critical depends on alpha and sided.

This function inverts the formula in two ways:

1. Solves for cluster_fraction given bulk_se, sc_zscore, and condition_fraction
2. Solves for bulk_se given cluster_fraction, sc_zscore, and condition_fraction

If the required cluster_fraction exceeds 1 (unattainable), the function returns NA and includes a warning message.

Value

A list with class "required_design" containing:

`target_di`	Requested diagnosability index
`achieved_di_at_fixed_se`	DI achieved if solving for cluster_fraction
`required_cluster_fraction`	Cluster fraction needed to reach target DI (at fixed bulk_se)
`required_bulk_se`	Bulk SE needed to reach target DI (at fixed cluster_fraction)
`achieved_di_at_fixed_cf`	DI achieved if solving for bulk_se
`scenario_fixed_se`	Message describing the fixed-SE scenario
`scenario_fixed_cf`	Message describing the fixed-CF scenario
`warnings`	Character vector of warnings if targets are unattainable

Examples

```
# Example: k=5 tissue-only case study
# Current DI is only 0.065; what would we need for DI = 0.3?

design <- required_design(
  target_di = 0.3,
  sc_zscore = 2.077409,
  condition_fraction = 0.631,
  n_cluster = 1332,
  cluster_fraction = 1332 / 19175,
  bulk_se = 0.062774,
  alpha = 0.05,
  sided = "one"
)

print(design)
```

`summary.bulknull` *Summary method for bulknull objects*

Description

Displays a concise one-line summary of the diagnosis.

Usage

```
## S3 method for class 'bulknull'
summary(object, ...)
```

Arguments

`object`	A bulknull object
`...`	Additional arguments (ignored)